

NuSTAR Calibration Coffee Hour

MLI Updates

Hannah Earnshaw, 2026/04/14

What is MLI?

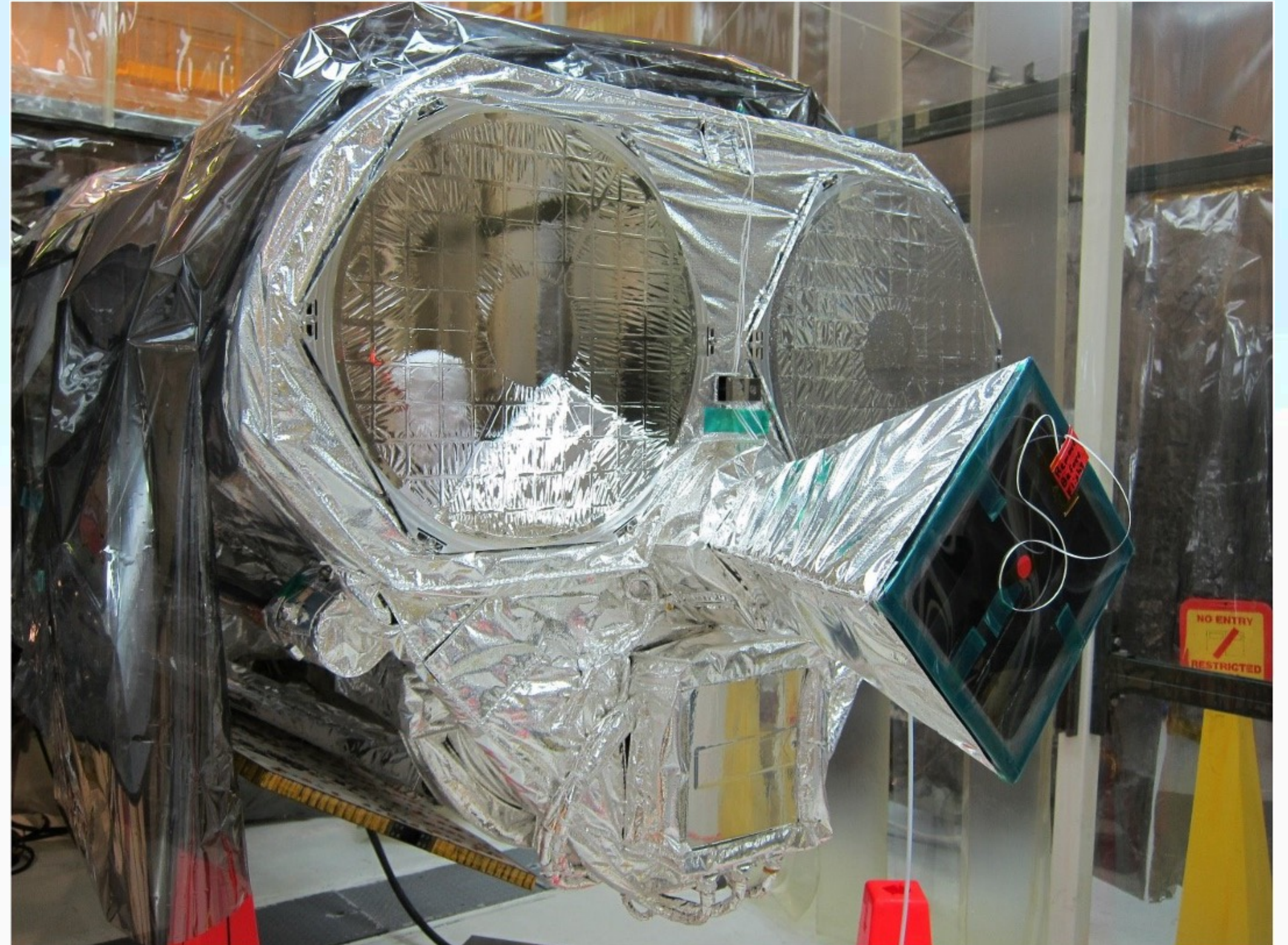
MLI = Multi-layer Insulation

Helps keep the NuSTAR optics thermally stable

Photo is NuSTAR pre-launch, with OPT1 closest to us and OPT0 to the right

OPT0 serves focal plane module FPMA, and OPT1 serves FPMB

Mostly transparent to X-rays, but there is some energy-dependent attenuation at low energies



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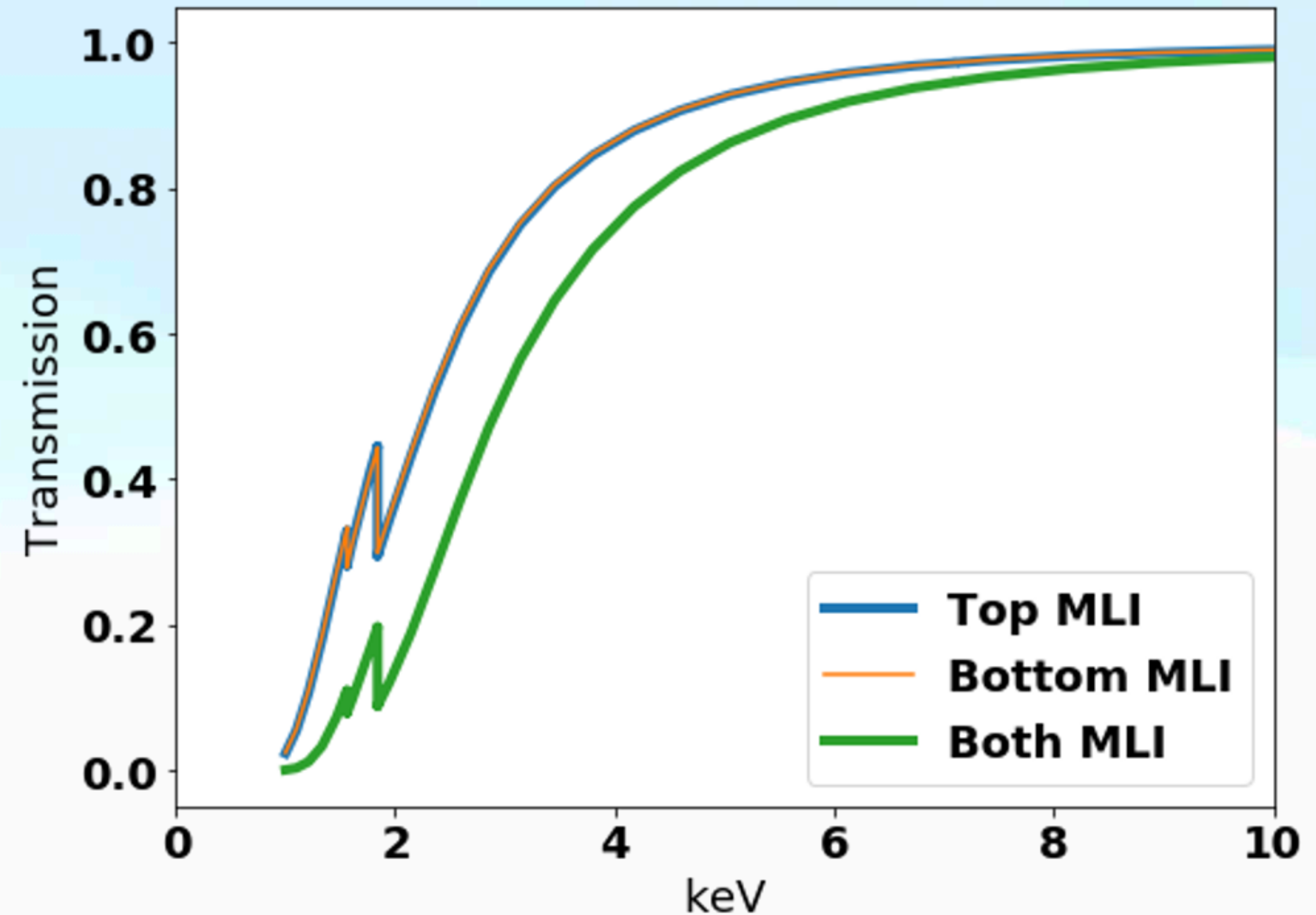
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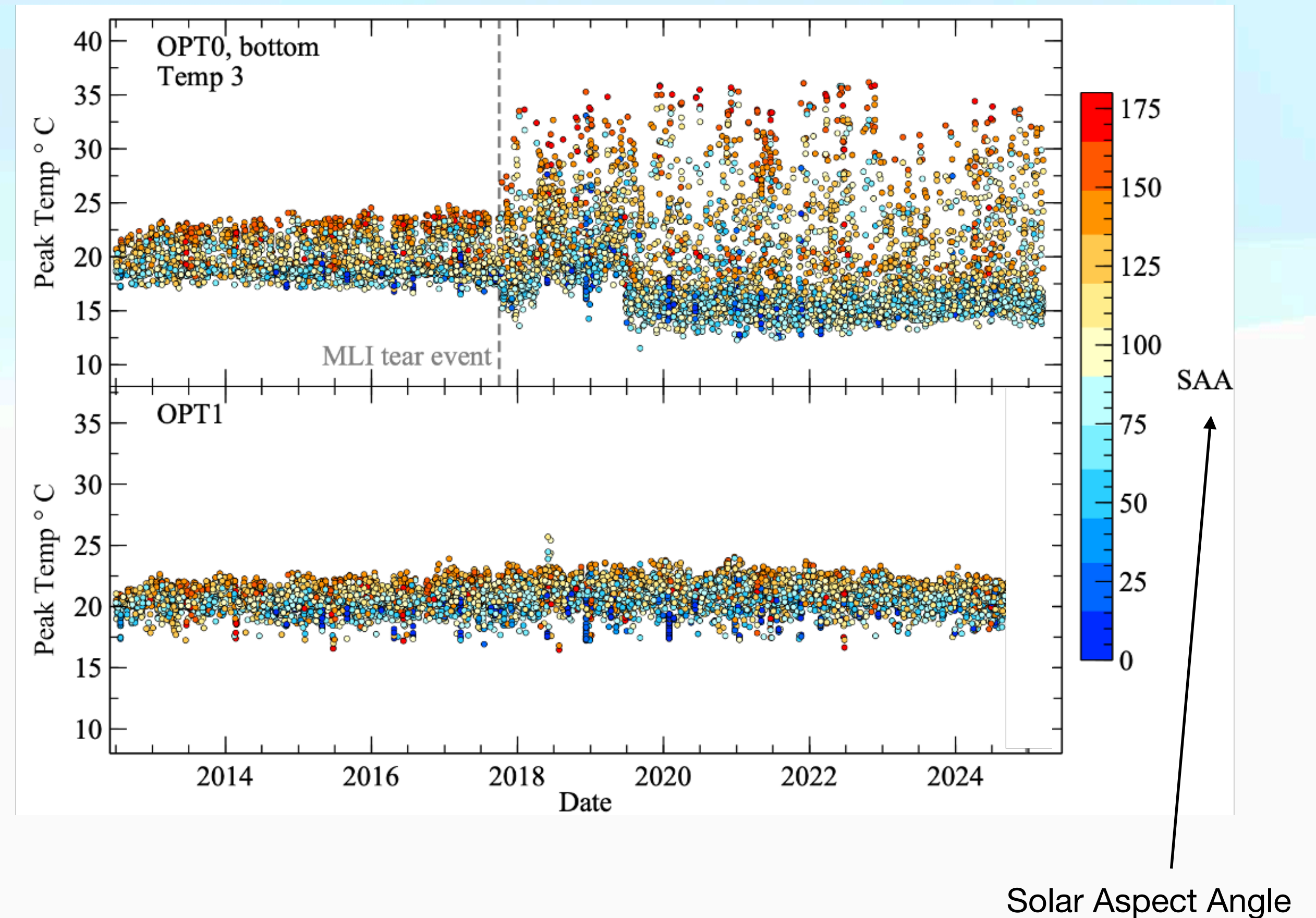
Finding a tear using temperature monitoring

Optics have six thermistors on each module, three on top and three on the bottom

Optics normally held at steady temperature of ~20C

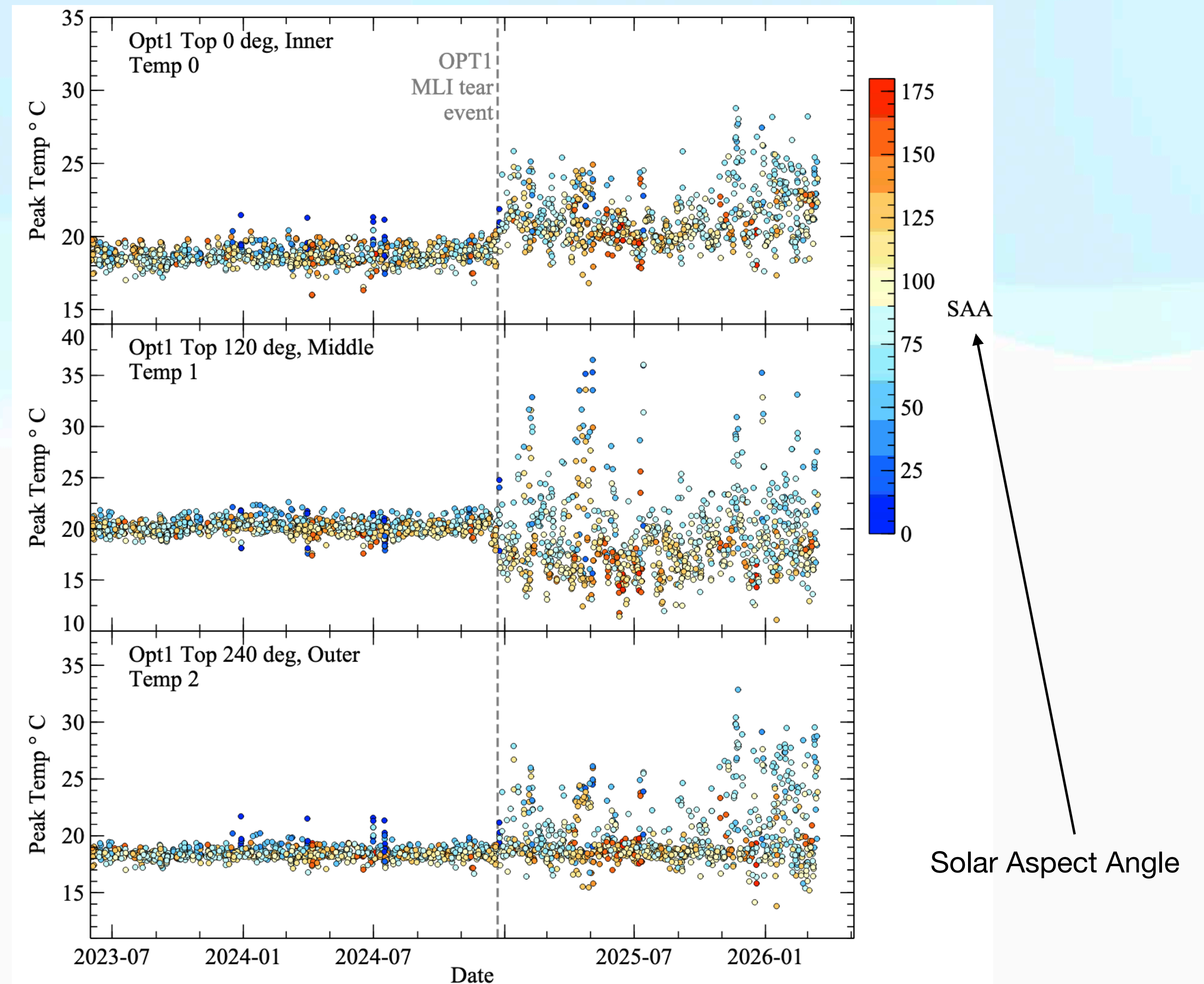
A sudden change in the temperature indicates that something has happened! Some parts are getting directly illuminated by the Sun, others are losing heat into space.

First tear event on OPT0 in late 2017 led to high temperatures measured by Temp 3, where optic was illuminated at high angles from the Sun



Finding a tear using temperature monitoring

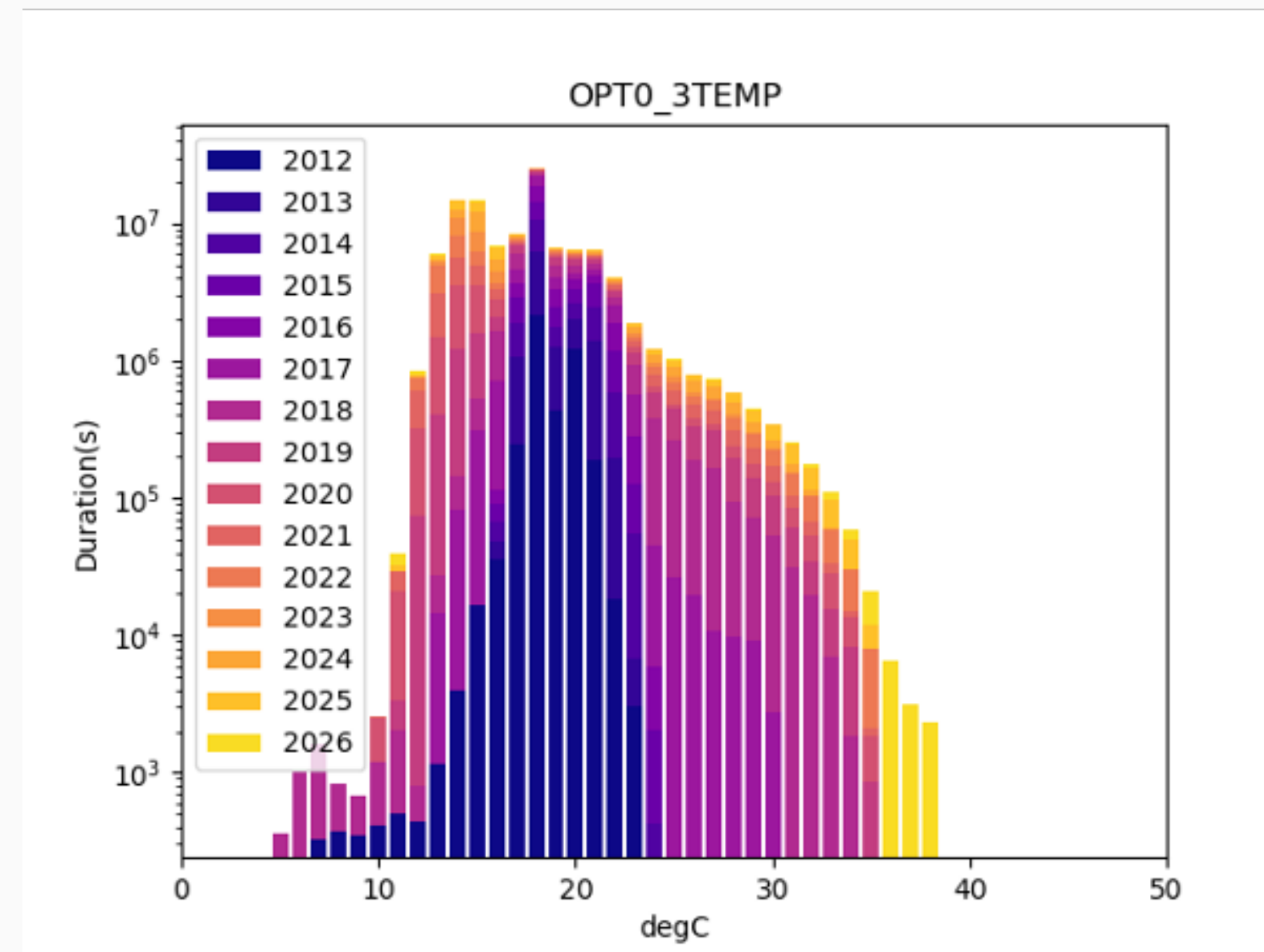
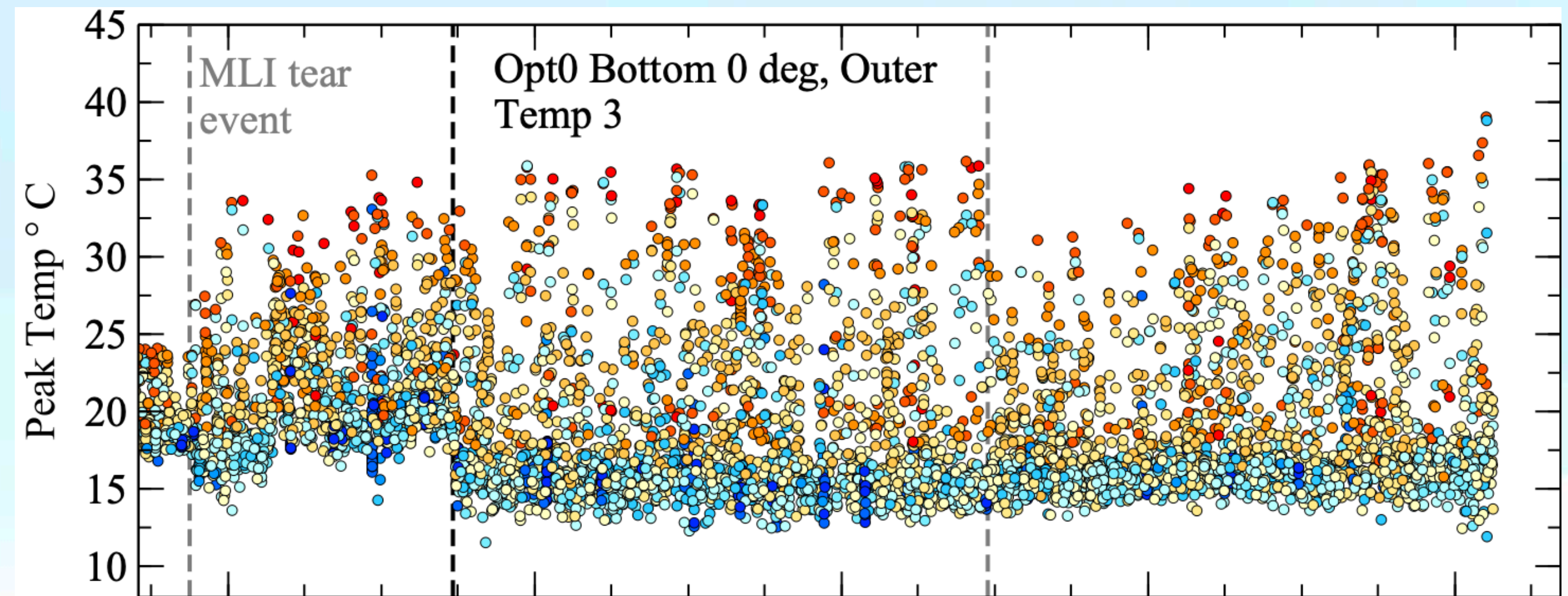
Second tear event happened in late December 2024 on the top of OPT1, leading to high temperatures seen by Temp 1 and Temp 2 when the spacecraft is pointing near the Sun and therefore illuminating the top of the optics.



Finding a tear using temperature monitoring

In 2026, the maximum temperature of the bottom of OPT0 has been rising, which may be due to additional tearing or worsening of the existing tear. While we don't expect the epoxy on the optics to deform at these temperatures, there is a cumulative effect that increases exponentially with temperature.

Scheduling Impacts: in order to avoid heating the optics to temperatures that may be dangerous to the epoxy, we have instated scheduling constraints to avoid observing targets with $SAA < 60$ and $SAA > 130$.



Effect on science

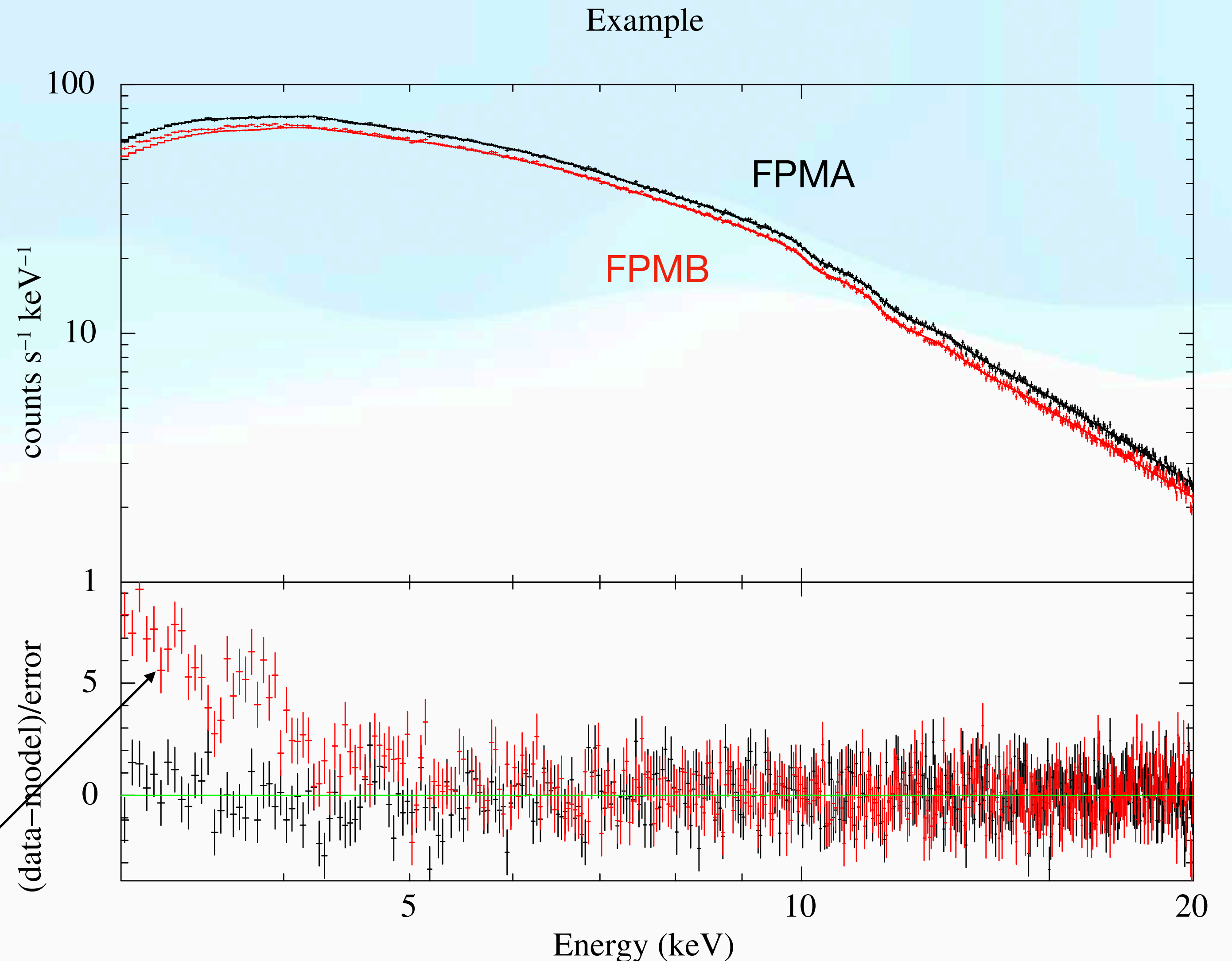
Data impacts: these tears lead to higher photon throughput below 7 keV that we need to correct for

The FPMA tear is steady at a fractional coverage of 0.82 (despite recent temperature increases) and is accounted for in the CALDB ARFs.

The increased FPMB throughput leads to an excess over FPMA. Its fractional coverage is around 0.9 but has been varying over time — no CALDB update yet. We are tracking this with regular Crab observations.

More info: Madsen (2020)
<https://arxiv.org/abs/2005.00569>

FPMB excess



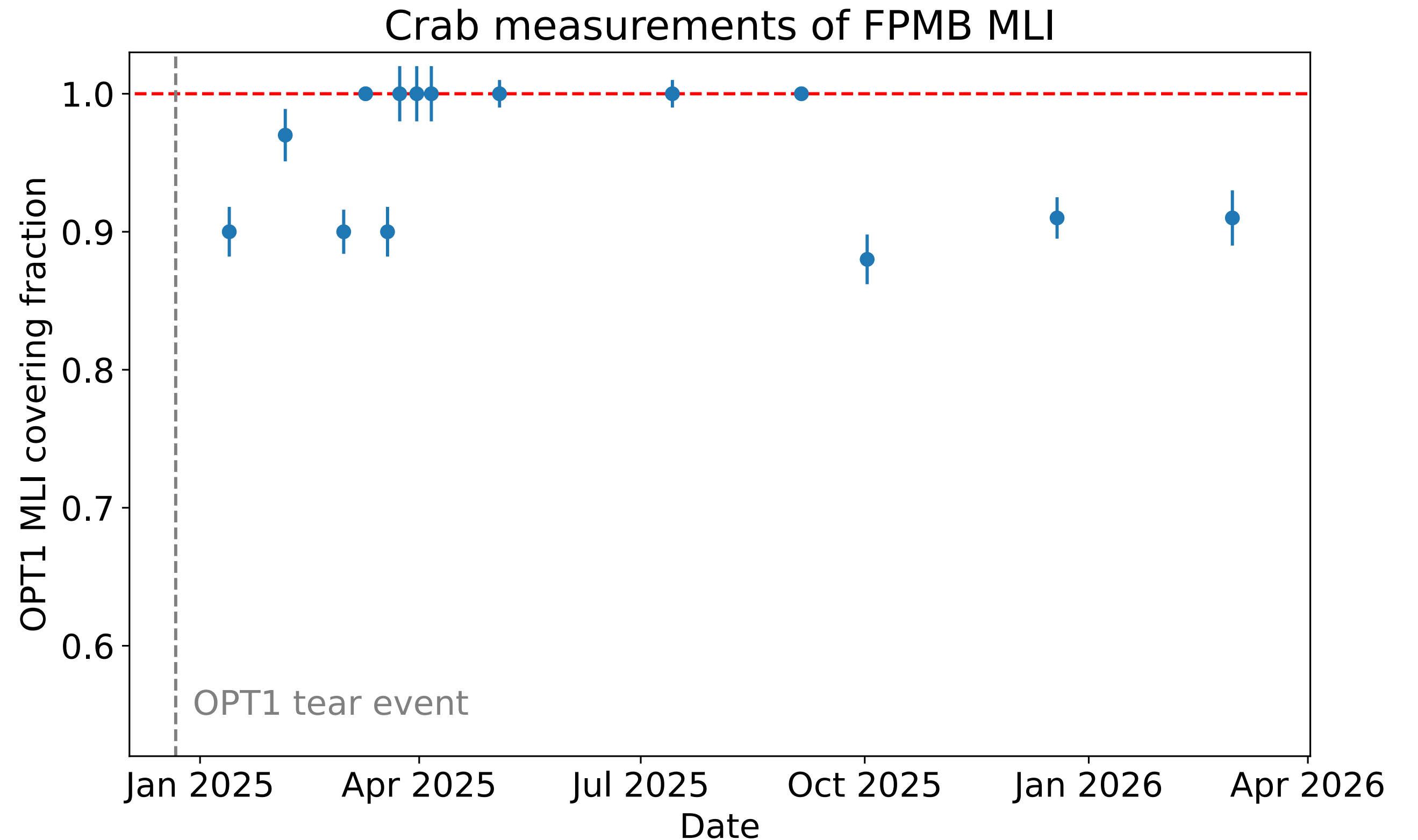
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What to do if your data is affected

CAVEAT

The following **only** applies to...

- **Bright** sources ($>\sim 100$ c/s) that were...
- Observed **after** 2024 which show...
- A clear discrepancy between FPMA and FPMB (increased FPMB) at energies < 7 keV

If you see this, let us know! We can help to confirm that MLI is the source of the discrepancy and advise on analysis

Obtain the MLI mtable model from the NuSTAR Operations Site here:

https://nustarsoc.caltech.edu/NuSTAR_Public/NuSTAROperationSite/mli.php

Available MLI models:

nuMLIv1.mod

Generated May 01, 2020

What to do if your data is affected

How to use the MLI mtable:

1. First, ignore FPMB between 3 and 7 keV when doing your original fits. Make sure the model includes a constant offset between FPMA and FPMB. Find the value of the constant and freeze it. Possibly also freeze NH, since it can be degenerate with MLI fraction.
2. Notice FPMB between 3 and 7 keV and add the mtable to the model. Set the value of the MLI fraction to 1.0 for FPMA and freeze it. Allow only the MLI fraction of FPMB to fit.
3. Please note: The range of MLI fraction may be between 0.6 and 1.0 but current observations suggest a value around 0.9 for FPMB. If you have any concerns about a particular observation, please contact the NuSTAR Helpdesk at the HEASARC.

```
XSPEC12>data 1:1 nuFPMAspec_grp.pha 2:2 nuFPMBSpec_grp.pha
XSPEC12>ignore bad
XSPEC12>ignore **0.-3. 79.-**
XSPEC12>ignore 2:3.-7.
```

```
XSPEC12>model const*tbabs*powerlaw
XSPEC12>fit
XSPEC12>freeze 2,5
```

=====						
Model	constant<1>	*TBabs<2>	*powerlaw<3>	Source No.: 1	Active/On	
Model	Model	Component	Parameter	Unit	Value	
par	comp					
Data group: 1						
1	1	constant	factor		1.00000	frozen
2	2	TBabs	nH	10^22	0.735897	frozen
3	3	powerlaw	PhoIndex		2.10713	+/- 9.38088E-04
4	3	powerlaw	norm		9.55044	+/- 1.92313E-02
Data group: 2						
5	1	constant	factor		0.978625	frozen
6	2	TBabs	nH	10^22	0.735897	= p2
7	3	powerlaw	PhoIndex		2.10713	= p3
8	3	powerlaw	norm		9.55044	= p4

```
XSPEC12>notice 2:3.-7.
XSPEC12>editmod const*tbabs*powerlaw*mtable{nuMLIv1.mod}

Input parameter value, delta, min, bot, top, and max values for ...
0.8 0.005( 0.008) 0 0 1 1
5:data group 1::fractionalMLI:MLI2_frac>1 -1

Input parameter value, delta, min, bot, top, and max values for ...
0.8 0.005( 0.008) 0 0 1 1
10:data group 2::fractionalMLI:MLI2_frac>1

XSPEC12>fit
```

=====						
Model	constant<1>	*TBabs<2>	*powerlaw<3>	*mtable{nuMLIv1.mod}<4>	Source No.: 1	Active/On
Model	Model	Component	Parameter	Unit	Value	
par	comp					
Data group: 1						
1	1	constant	factor		1.00000	frozen
2	2	TBabs	nH	10^22	0.735897	frozen
3	3	powerlaw	PhoIndex		2.10676	+/- 9.22952E-04
4	3	powerlaw	norm		9.53066	+/- 1.87343E-02
5	4	fractionalMLI	MLI2_frac		1.00000	frozen
Data group: 2						
6	1	constant	factor		0.978625	frozen
7	2	TBabs	nH	10^22	0.735897	= p2
8	3	powerlaw	PhoIndex		2.10676	= p3
9	3	powerlaw	norm		9.53066	= p4
10	4	fractionalMLI	MLI2_frac		0.813340	+/- 7.34013E-03

Going forwards

We are continuing to investigate thermal behavior with Solar aspect angle. While our new scheduling constraints will hopefully mean we avoid very high temperatures, there may still be changes in MLI coverage.

Without OPT1 as a stable baseline, modeling the evolution of the MLI in both optics at once is a little more difficult (but not impossible!) We will be continuing to monitor the Crab and other bright sources regularly to keep track of this.

The FPMA ARF remains steady. We may release a CALDB update for the FPMB ARF if the MLI stabilizes, but in the meantime a correction may be required for bright sources.

Calibration Coffee slides available at: https://github.com/NuSTAR/calibration_coffee